WP2 Solutions

125. $u = \langle 2,2,-1\rangle$, $v = \langle -1,2,2\rangle$

$$u\cdot v = 2(-1)+2(2)+(-1)2$$
$$= -2+4-2$$
$$= 0$$

145. a orthogonal to $b = \langle 3,4\rangle$

let $a = \langle x,y\rangle$ then
$a\cdot b = 3x+4y$
and $a\cdot b = 0$
so $3x+4y=0$
or $y = -\frac{3}{4}x$
so $a = \langle x, -\frac{3}{4}x\rangle$ for $x \in \mathbb{R}$

171. $u = 4\hat{i} - 3\hat{j}$, $v = 3\hat{i} + 2\hat{j}$

A) $w = \text{proj}_u v$

$$= \frac{4(3)+(-3)(2)}{4^2+(-3)^2}\langle 4,-3\rangle$$
$$= \frac{12-6}{16+9}\langle 4,-3\rangle$$
$$= \left\langle \frac{24}{25}, \frac{-18}{25}\right\rangle$$

B) $q = \langle 3,2\rangle - \left\langle \frac{24}{25}, \frac{-18}{25}\right\rangle$
$= \left\langle \frac{51}{25}, \frac{68}{25}\right\rangle$

so $v = \left\langle \frac{24}{25}, -\frac{18}{25}\right\rangle + \left\langle \frac{51}{25}, \frac{68}{25}\right\rangle$

131. $a = \langle 3,-1\rangle$, $b = \langle -4,0\rangle$

A) $a\cdot b = 3(-4)+(-1)(0) = -12$
$\|a\| = \sqrt{3^2+(-1)^2} = \sqrt{10}$
$\|b\| = \sqrt{(-4)^2+0^2} = 4$

$$\cos\theta = \frac{-12}{\sqrt{10}(4)}$$
$$\theta = \cos^{-1}\left(\frac{-3\sqrt{10}}{10}\right)$$
$$\theta = 2.82 \text{ rad}$$

B) θ is not acute.

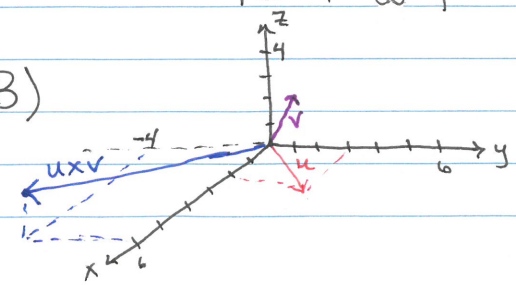
177. $\|F\| = 25$, $\|\overrightarrow{PQ}\| = 50$
and $\theta = 20^\circ$

$$W = 25(50)\cos 20^\circ$$
$$= 1175 \text{ ft}\cdot\text{lbs}$$

185. $u = 2\hat{i}+3\hat{j}$, $v = \hat{j}+2\hat{k}$

A) $u\times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 0 \\ 0 & 1 & 2 \end{vmatrix} = 6\hat{i}-4\hat{j}+2\hat{k}$
or $\langle 6,-4,2\rangle$

B)



189. $u = \langle 3,-1,2\rangle$, $v = \langle -2,0,1\rangle$

$$u \times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 2 \\ -2 & 0 & 1 \end{vmatrix}$$

$$= (-1-0)\hat{i} - (3+4)\hat{j} + (0-2)\hat{k}$$

$$= \langle -1,-7,-2\rangle$$

$$\|u \times v\| = \sqrt{(-1)^2 + (-7)^2 + (-2)^2}$$

$$= \sqrt{54}$$

$$= 3\sqrt{6}$$

$$w = \frac{u \times v}{\|u \times v\|} = \left\langle \frac{-\sqrt{6}}{18}, \frac{-7\sqrt{6}}{18}, \frac{-\sqrt{6}}{9} \right\rangle$$

199. $u = \langle -1,0,e^t\rangle$, $v = \langle 1,e^{-t},0\rangle$

$$w = u \times v$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & e^t \\ 1 & e^{-t} & 0 \end{vmatrix}$$

$$= (0-e^0)\hat{i} - (0-e^t)\hat{j} + (-e^{-t}-0)\hat{k}$$

$$= -\hat{i} + e^t\hat{j} - e^{-t}\hat{k}$$

or $\langle -1, e^t, -e^{-t}\rangle$

209. $u = \langle 3,2,0\rangle$ $v = \langle 0,2,1\rangle$

$$u \times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 0 \\ 0 & 2 & 1 \end{vmatrix}$$

$$= (2-0)\hat{i} - (3-0)\hat{j} + (6-0)\hat{k}$$

$$= \langle 2,-3,6\rangle$$

$$A = \|u \times v\|$$

$$= \sqrt{2^2 + (-3)^2 + 6^2}$$

$$= 7$$

235. $\|r\| = 12$ in $F = 10$ lbs $\theta = 60°$

$$\|\tau\| = 1(10)\sin 60°$$

$$= 8.66 \text{ ft}\cdot\text{lb}$$

237. $\|\tau\| = 100$ N·m, $\|r\| = 20$cm, $v = \langle 0,1,-2\rangle$

$$u = \langle 0,1,0\rangle \Rightarrow \cos\theta = \frac{0(0) + 1(1) + 0(-2)}{\sqrt{0^2+1^2+0^2}\sqrt{0^2+1^2+(-2)^2}}$$

$$\theta = \cos^{-1}\left(\frac{\sqrt{5}}{5}\right)$$

$$100 = 0.2\|F\|\sin\left(\cos^{-1}\left(\frac{\sqrt{5}}{5}\right)\right)$$

$$\|F\| = \frac{100}{0.2\sin\left(\cos^{-1}\left(\frac{\sqrt{5}}{5}\right)\right)}$$

$$\|F\| = 559 \text{ N}$$